

Topic:-

Take two plastic glasses and a thread between them to make a toy phone. Discuss your conclusions.

Introduction:-

Sound is an essential form of energy that produces the sensation of hearing in our ears and allows living beings to communicate. In our daily lives, we hear chirping birds, musical instruments, roaring vehicle horns, and human conversations. In Chapter 15 of our Class 8 Science textbook, we learn that sound is always produced exclusively by vibrating bodies.

When we speak, our vocal cords vibrate. For sound waves to travel from the speaker to the listener's ear, a material medium is strictly required—sound cannot travel through a vacuum! While we usually hear sound traveling through atmospheric air (gas), sound can also travel through liquids (water) and solids (string, metal, wood). In fact, because molecules in dense solids are packed closely together, sound travels much faster and clearer through solid objects than through air. In this project, we constructed a simple toy telephone using two plastic glasses and a stretched cotton thread to test acoustic transmission through solids.

❖ Speed of Sound Across Different Mediums:-

- **In Gases (Air at 20°C):** Molecules are loosely spaced; sound travels at ~340 m/s.
- **In Liquids (Water at 20°C):** Molecules are closer; sound travels at ~1,480 m/s.
- **In Solids (Cotton String / Wood):** Molecules are tightly bound; sound travels at over 3,000 m/s!

Aim & Craft Materials Required:-

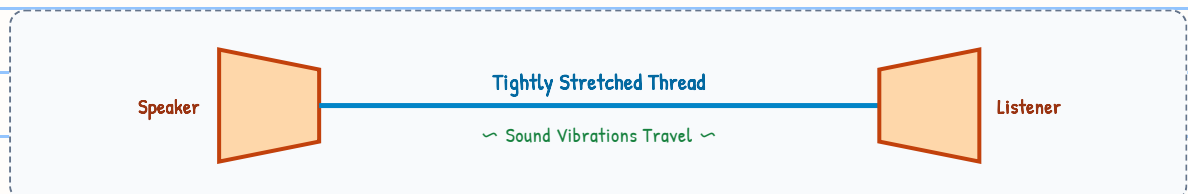
Aim: To make a working toy telephone using two plastic glasses and a connecting thread, test sound transmission under different tension states, and discuss our conclusions.

❖ *Objectives of the Experiment:-*

- To construct a functional acoustic string telephone using low-cost materials.
- To verify that solid string conducts mechanical sound vibrations over long distances.
- To examine how string tension (tight vs loose) and finger damping affect voice clarity.

❖ *Materials Required:-*

- Two identical disposable plastic glasses (or paper cups).
- 10 meters of strong cotton sewing thread (and alternative nylon fishing line).
- Two small pieces of wooden matchsticks (or metal paper clips), sewing needle, scissors.



Step-by-Step Construction Procedure:-

We constructed and tested our toy telephone following these straightforward steps:

- **Step 1: Piercing Cup Bases:** Using a sharp pin, we pierced a neat tiny pinhole right in the center of the circular bottom of each plastic glass.
- **Step 2: Threading the Line:** Pushed one end of the 10-meter cotton thread through the bottom hole from outside into the cup.
- **Step 3: Knotting with Matchstick:** Tied the inner end securely around a small cut piece of wooden matchstick so the thread could not slip back through the hole.
- **Step 4: Connecting the Second Cup:** Repeated the identical threading and knotting process with the second cup at the other end of the thread.
- **Step 5: Testing with Partners:** Two students walked apart in the school corridor until the thread was pulled completely straight and taut without touching any walls or floors.
- **Step 6: Voice Transmission:** One student spoke in a soft whisper into one cup, while the second student held the other cup firmly over their ear to listen.

Observations & Acoustic Test Table:-

We tested different experimental conditions and recorded whether sound was heard clearly:

<i>Test Condition</i>	<i>State of Thread</i>	<i>Can Listener Hear Whisper?</i>	<i>Scientific Physical Reason</i>
Straight & Taut	Pulled completely tight	Yes, crystal clear & loud!	Tension transfers longitudinal wave pulses
Loose / Sagging	Slack; hangs down loosely	No, very faint or silent	Loose string sags and absorbs kinetic energy
Pinched with Fingers	Middle held firmly by hand	Sound cuts off instantly!	Fingers absorb and damp the vibrations
Touching Wall Corner	Bent over brick wall edge	Sound becomes muffled	Wall surface absorbs vibration energy
Nylon vs Cotton Thread	Taut nylon fishing wire	Even louder and crisper!	Dense elastic nylon transmits waves with less loss

Scientific Analysis & Discussion:-

Analyzing the physics of our toy telephone explains how mechanical waves propagate:

- **The Cup Acting as a Diaphragm:** When the speaker whispers into the cup, voice sound waves strike the thin, flexible plastic bottom. The plastic base acts as an acoustic diaphragm, vibrating rapidly back and forth.
- **Transmission Along the Thread:** These vibrations pull and release the tightly stretched cotton thread, launching rapid longitudinal compression-rarefaction wave pulses along the string molecules at over 3,000 meters per second.
- **Receiver Diaphragm Action:** At the opposite end, the vibrating string pulls and pushes the bottom of the listener's cup. The cup base vibrates the air column inside the cup, creating sound waves that strike the listener's eardrum!

❖ Why String Must Be Kept Taut:-

When the string is loose, the mechanical pulses cannot pull the next molecule forward; instead, the string merely bends and sags, dissipating all sound energy.

Real-World Connections & Historical Context:-

Our simple toy phone represents foundational principles used in telecommunications:

- **Robert Hooke's Acoustic Phone (1667):** English polymath Robert Hooke invented the first string telephone over 350 years ago, demonstrating long-distance sound conduction through stretched wire.
- **Doctors' Stethoscope:** A physician's stethoscope operates on the exact same acoustic principle. A chest diaphragm captures heartbeat sounds and transmits them through solid rubber tubes directly into the doctor's ears.
- **Alexander Graham Bell's Electric Telephone (1876):** Bell replaced the mechanical vibrating string with an electric wire: sound waves vibrated a metal diaphragm, varying electrical resistance, which was turned back into sound at the receiving earphone!

❖ Fun School Activity Variation:-

Connecting a third cup to the center of the main line creates a 3-way party line where three students can talk simultaneously!

What We Learned & Conclusion:-

- Sound is produced by vibrating bodies and requires a material medium for propagation.
- Sound travels efficiently through dense solids (cotton thread) faster and clearer than through air.
- Connecting thread must be kept under tension (taut) so longitudinal vibrations can pass without damping.
- The plastic cup bottoms act as transmitter and receiver diaphragms, mimicking modern microphones and speakers.

Conclusion:-

Our plastic cup toy telephone was an outstanding practical physics demonstration. Whispers that were inaudible through air across a 10-meter corridor were heard loud and clear through the tight string. This simple experiment proved that sound travels effectively through solids via mechanical vibrations, providing a delightful hands-on introduction to the science of acoustic communication.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 15: Sound).
- Hands-On Physics Experiments for Middle School, Balbharti Pune.
- NCERT Class 8 Science — Sound and Wave Propagation.